

Class XI | Unit 3 | weight 15%

Inside the system unit

Board weight about 15% | 10 theory periods and 10 lab periods

The system unit is the case that holds the working parts of a desktop computer. This lecture opens the case in words: the board, the processor, the buses, the ports and the slots.

What you should be able to do

- Describe CPU casings and the main parts inside the system unit.
- Name the components on the motherboard and the job of each.
- Explain the internal architecture and the external pins of a microprocessor.
- Read a processor specification.
- Define address, data and control buses and say which way data flows.
- Describe serial, parallel, USB and HDMI ports, and PCI, EISA and VESA slots.

Path through the lecture

- 1 Point at a real case before drawing the motherboard.
- 2 Teach the processor as one chip with a clear inside and a clear outside.
- 3 Use one numerical example of an address bus.
- 4 Finish in the lab by seating RAM and identifying one port.

3.1 The system unit

System unit. The case that houses the motherboard, processor, memory, storage, power supply, ports and cooling of a desktop computer.

Cases differ by size and by how easily they can be expanded. A **tower** stands on the floor, has room for drives and cards, and the fans can move air through a tall space. A **desktop** case lies flat and can hold a monitor, with less room to expand. A **small-form** case saves desk space and accepts fewer cards. An **all-in-one** builds the computer into the monitor housing, which is tidy and harder to upgrade.

- The **power supply** converts mains AC into the low-voltage DC the parts use, and it feeds them through labelled connectors.
- The **motherboard** is the main circuit board. The processor, memory and cards are mounted on it or plugged into it.
- The **hard drive** stores files. A **combo drive** reads optical discs.
- **Ports** are the sockets on the edge of the case. The **cooling system** is the heat sink and fan on the processor, plus the fans that move air through the case.

3.2 The motherboard

Every important signal of the computer meets on the motherboard. Learn the part and the job together.

Parts on the motherboard

Part	Function
Microprocessor	Executes instructions. It sits in a socket on the board.
BIOS in ROM	Permanent start-up firmware. It runs the power-on self-test and starts loading the operating system.

Part	Function
RAM slots	Hold the memory modules that form main memory.
CMOS battery	Keeps the clock and the saved setup when the computer is unplugged.
Jumpers	Small pins and clips that select a setting, used more on older boards.
Ports	Connect external devices such as a keyboard, a display or a USB drive.
Expansion slots	Accept cards that add a function, such as a better display or a network.
System bus	The shared paths that carry addresses, data and control signals among the parts.

3.3 The microprocessor

Microprocessor. A processor built on a single integrated circuit. In a PC it is the CPU.

Inside the chip, five ideas explain how it works. The **clock** sends a regular pulse that times every step. A 3 GHz clock ticks three thousand million times a second. **Registers** are tiny storage locations inside the processor, faster than RAM, used for the instruction being executed, the address being used, and the current result. The **ALU** performs arithmetic and logical operations. The **control unit** fetches an instruction from memory, decodes it, and tells the other parts when to act. **Cache** is a small SRAM store of recently used instructions and data, so the processor waits less often for DRAM.

Outside, the chip is known by its pins. **Address pins** carry the address of the memory location the processor wants. They carry that address outward, so the address bus is unidirectional. **Data pins** carry the value being read or written, so they are bidirectional. **Control pins** carry signals such as read, write, and whether the address refers to memory or to an input-output device. **Interrupt pins** let a device ask the processor to pause and attend to it. **Power pins** supply voltage and ground.

A specification names the manufacturing technology, often in nanometres, the generation or family, the clock speed, the word size in bits (32 or 64), the width of the data bus, and the cache sizes at each level. A wider data bus moves more bits in one transfer. A wider address bus can name more locations: a 16-bit address bus can name 2 to the power 16, which is 65536 locations.

3.4 Buses

Bus. A shared set of electrical lines that carries signals from one component to another.

The system bus

Bus	Carries	Direction
Address bus	The address of a memory location or a port	Unidirectional, from the processor
Data bus	The data being read or written	Bidirectional
Control bus	Timing and command signals such as read, write and interrupt	The separate lines run in the direction each signal needs

Unidirectional means the signal travels one way. Bidirectional means it can travel either way, not both ways on the same wire at the same instant. The address leaves the processor. Data may leave or enter.

3.5 Ports

- A **serial port** sends bits one after another on a single data line. The older COM port is the classroom example. The cable can be longer. The transfer is slower than a wide parallel cable of the same era.
- A **parallel port** sends a group of bits at the same time, one bit on each wire. The older printer port is the example. It is fast over a short cable.

- **USB**, Universal Serial Bus, is the general port on a modern PC. Devices can be plugged in while the computer is on. One port can feed a hub of several devices, and the bus can supply power to small devices. Later versions of USB move data faster.
- **HDMI**, High-Definition Multimedia Interface, carries digital video and audio together to a monitor, a television or a projector.

3.6 Expansion slots

PCI, Peripheral Component Interconnect, is the common slot used to add a card such as a network card or a sound card. **EISA**, Extended Industry Standard Architecture, is an older 32-bit slot that also accepted older ISA cards. The curriculum's third name, written VISA in the acronym list, refers in this topic to the **VESA local bus** slot. That slot was tied closely to the processor bus so a video card could move data quickly. It belongs to the generation of PCs before PCI became the usual choice.

Exam point

If the question lists PCI, EISA and VISA, describe three expansion slots. In the answer you may write VESA local bus and add that the curriculum acronym is VISA. Describe the slot, not Virtual Instrument Software Architecture, which is a different use of those letters.

In the lab

Lab safety: shut the computer down, unplug the mains, and press the power button once to discharge the supply. Seat RAM only in the matching slot, with the notch lined up, until the clips close. Do this with your teacher present.

Key terms

Term	Meaning to write in an answer
Motherboard	The main circuit board that connects the processor, memory, storage and cards.
ALU	The part of the CPU that performs arithmetic and logic.
Control unit	The part of the CPU that fetches, decodes and directs instructions.
Register	A small, fast storage location inside the CPU.
Cache	Fast SRAM that holds recently used instructions and data.
Bus	A shared path of wires carrying address, data or control signals.
Port	A socket used to connect an external device.
BIOS	Firmware in ROM that starts the computer.

Bring it together

- The case, power supply, motherboard, drives, ports and cooling make the system unit.
- The processor clock, registers, ALU, control unit and cache explain the inside. Address, data, control, interrupt and power pins explain the outside.
- Address lines leave the processor. Data lines carry values in or out.
- Serial, parallel, USB and HDMI are ports. PCI, EISA and the VESA local bus are expansion slots.

Do this in the practical

- 1 Identify the processor socket, RAM slots, CMOS battery, expansion slots and one USB port on a motherboard.
- 2 With the power unplugged and a teacher present, remove and reseat a RAM module. Note the notch and the clips.

- 3 Enter the BIOS or UEFI setup and find the boot-order screen. Leave the setting as you found it unless the teacher asks for a change.

Check yourself

Attempt every item before you read the key. Section A of the Computer Science paper is a set of multiple-choice questions.

1. The ALU is responsible for:

- A. Arithmetic and logic
- B. Long-term file storage
- C. Converting AC to DC
- D. Printing

2. Which bus carries the location the processor wants to read?

- A. Data bus
- B. Address bus
- C. HDMI
- D. Power supply

3. The address bus is:

- A. Bidirectional
- B. Unidirectional from the processor
- C. A printer only
- D. A type of ROM

4. Cache memory is usually built from:

- A. DRAM
- B. SRAM
- C. CD-R
- D. Magnetic tape

5. The CMOS battery:

- A. Prints documents
- B. Keeps setup and the clock when mains power is off
- C. Is the main hard disk
- D. Translates C++

6. HDMI carries:

- A. Digital video and audio
- B. Only mains electricity
- C. Only a paper signal
- D. Ultraviolet light

7. A 10-bit address bus can name:

- A. 10 locations
- B. 100 locations
- C. 1024 locations
- D. 8 locations

8. PCI is a:

- A. Type of RAM
- B. Expansion slot
- C. Keyboard layout
- D. File attribute

Answer key

Q	Answer	Why this is the answer
1	A	The arithmetic and logic unit calculates and compares.
2	B	The address bus carries the address of the location.
3	B	Addresses are issued by the processor toward memory.
4	B	Cache needs the speed of SRAM.
5	B	The battery maintains saved setup information and the time.
6	A	HDMI is a digital multimedia port for picture and sound.
7	C	Ten address lines name 2 to the power 10, which is 1024 locations.
8	B	PCI is an expansion slot for add-in cards.

Short answers

A 3-mark answer needs a definition and two clear points, or a three-row difference. Write in sentences.

Q1. Write the functions of the ALU and the control unit.

Model answer

The ALU performs arithmetic operations such as addition and logical operations such as comparison. The control unit fetches each instruction from memory, decodes it, and directs the other parts of the computer to carry it out.

Q2. Differentiate address bus and data bus.

Model answer

The address bus carries the address of a memory location and is unidirectional, from the processor. The data bus carries the data being read or written and is bidirectional. The width of the address bus limits how many locations can be named. The width of the data bus limits how many bits move in one transfer.

Q3. What is the BIOS?

Model answer

The BIOS is firmware stored in ROM on the motherboard. When the computer is switched on it runs a self-test and starts the process of loading the operating system.

Q4. Why is a heat sink fitted on the processor?

Model answer

The processor produces heat while it works. The heat sink, with its fan, conducts that heat away so the chip stays within a safe temperature.

Q5. Give one use of USB and one use of HDMI.

Model answer

USB connects general devices such as a pen drive, a printer or a keyboard, and it can supply power to small devices. HDMI connects a display and carries digital video and audio together.

Detailed answers

Section C asks for a connected answer. Open with a definition, develop the points below, and close with one example.

Question 1. Describe the main components of the system unit and the main parts found on a motherboard.

- 1 Define the system unit and name tower, desktop and all-in-one as case styles.
- 2 Explain power supply, motherboard, hard drive, combo drive, ports and cooling.
- 3 On the motherboard, give the job of the processor, BIOS, RAM slots, CMOS battery, ports, expansion slots and the system bus.
- 4 Close with the path of a program: it is stored on the drive, loaded into RAM, and executed by the processor.

Question 2. Explain the internal parts and the external pins of a microprocessor.

- 1 Define a microprocessor.
- 2 Explain clock, registers, ALU, control unit and cache.
- 3 Explain address, data, control, interrupt and power pins.
- 4 Add one specification sentence: clock speed, bit size, bus width and cache are how two processors are compared.
- 5 Give the 16-bit address example: 65536 locations.